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THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF COMPUTING

EXAMINATION

Subject : COMP4442 Service and Cloud Computing

Session : 2023 / 2024 Semester II

Date : 10 May 2024 Time : 12:30 - 14:30

Time Allowed: 2 Hours

Subject Examiner(s): Dr ZHANG Zhaorui

This question paper has a total of ___5___ pages (cover included).
(Some pages may be intentionally omitted.)

Instructions to Candidates:

- This is an open-book exam.
- The exam is 100 marks in total.
- Any student caught cheating, e.g., sharing or copying answers with classmates, will receive a ZERO mark for the exam.

Do not turn over the page until you are told to do so!

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QUESTION 2 (15 MARKS)

Short Questions:

(1) What is the virtualization technique? (3 marks)

(2) List 2 virtualization techniques and describe their features, like Xen, Docker, etc. (3 marks)

(3) List 2 Data Center Network architectures and describe their features, like Fat-Tree and DCell. (3 marks)

(4) Please describe the relationship between Hadoop and Spark. (3 marks)

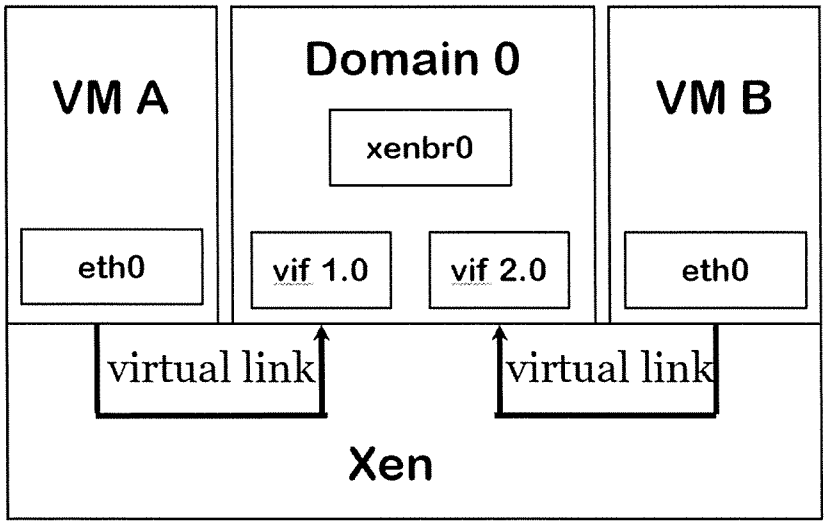
(5) What is the difference between a "transformation" and an "action" in Spark? (3 marks)

QUESTION 3 (15 MARKS)

Network virtualization is a transformative technology that abstracts and disjoints traditional networking functions from the underlying hardware, thus enabling the creation of multiple virtual networks on a shared physical infrastructure. Network virtualization software allows network administrators to move virtual machines across different domains without reconfiguring the network. The software creates a network overlay that can run separate virtual network layers on top of the same physical network fabric:

a) For network virtualization in the Xen virtualization technique, please describe how it works. (Hint: network interfaces, etc.) (8 marks)

b) Please figure out the data flow when virtual machine A (VM A) wants to send data to virtual machine B (VM B) in the following figure. (7 marks)



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QUESTION 4 (20 MARKS)

A hypervisor, also known as a virtual machine monitor (VMM), is a type of software that enables multiple virtual machines to run on a single physical server or host. There are two main types of hypervisor: Type-1 and Type-2. Please answer the following questions:

a) Xen is an open-source hypervisor technology that allows multiple virtual machines to run on a single physical machine, and it runs directly on the host machine's hardware and provides a layer of virtualization that allows multiple operating systems to run independently on the same physical machine.

   According to the above description, which type of virtualization that Xen belongs to, Type-1 or Type-2? (4 marks)

b) Please provide the main difference between Type-1 and Type-2 hypervisors, along with their strengths and weaknesses. (8 marks)

c) Kubernetes is an open-source container orchestration platform that automates the deployment, scaling, and management of the containerized application. It is often used for the management and scheduling of the container launched by Docker, etc. Such virtualization technology is also called containerization, an OS-level virtualization. Please explain what is "OS-Level" virtualization, along with its advantages compared to other virtualization techniques. (8 marks)

QUESTION 5 (20 MARKS)

Kubernetes, also known as K8S, is an open-source platform for managing containerized workloads and services. It was originally designed by Google and is now maintained by the Cloud Native Computing Foundation.

a) Previously, Docker is often used to create the container for Kubernetes. For the Docker, please describe what the "Cgroups" and "NameSpaces" are. (5 marks)
b) Explain the architecture of Kubernetes. (7 marks)
c) Describe the main workflow of Kubernetes and also draw a figure to show it. (8 marks)

QUESTION 6 (20 MARKS)

A Spark Python program based on RDDs is a program that uses the PySpark API to perform distributed computing tasks on RDDs in Python. The program typically follows a series of steps, including loading data from a data source, transforming the data using RDD operations like map, filter, and reduceByKey, and finally writing the results to an output storage system. Spark Python programs based on RDDs are typically used for processing and analyzing large-scale data sets in a distributed and fault-tolerant manner.

You are given a dataset, sales_data.csv, which contains sales records from a retail company. The dataset has the following schema:

order_id: Unique identifier for each sale (string)
product_id: Unique identifier for each product (string)
date: Date of the sale (string, format yyyy-MM-dd)
quantity: Number of units sold (integer)
price: Price per unit (float)

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We give the following Spark Code and please answer the following questions.

Spark Code:

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import col, sum, to_date

spark = SparkSession.builder.appName("Sales Data Analysis").getOrCreate()

sales_df = spark.read.csv("sales_data.csv", header=True, inferSchema=True)

sales_df = sales_df.withColumn("date", to_date(col("date"), "yyyy-MM-dd"))

sales_2024_df = sales_df.filter(col("date").between("2024-01-01", "2024-4-18"))
revenue_2024_df=sales_2024_df.groupBy("product_id").agg(sum(col("quantity")*
col("price")).alias("total_revenue")).orderBy(col("total_revenue").desc())

revenue_2024_df.show()

top_3_products_2024 = revenue_2024_df.limit(3)
top_3_products_2024.show()


spark.stop()
```

Please answer the following questions based on the above description, dataset, and Spark Code:

a) Please provide a description for each line of the code based on the knowledge of Spark and RDD. (5 marks)
b) What is the "Transformation" in Spark? Describe the "Transformation" and provide the example from the above Spark code. (5 marks)
c) What is the "Action" in Spark? Describe the "Action" and provide the example from the above Spark code. (5 marks)
d) What are the main components (architecture) of the Spark application, and explain each component. (5 marks)

END